Threshold Entry in a Fixed-Prize Contest: Measurement Map

Companion note to the proofs document

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The model is a static, one-shot contest for one prize of value V among Q wealth-heterogeneous challengers and one incumbent. Each challenger chooses $e_i \in \{0,1\}$, paying the fee c to enter; scores $\sigma_i = \mu r_i + (1-\mu)s_i e_i$ are then drawn, and the highest score takes the prize. The realised wealth profile is common knowledge, so the model contains no beliefs, priors or posteriors, and no observer-side parameter such as measurement precision. The discrete objects are Q, the incumbent's label $C \in \{F,G\}$, the decisions e_i, the entrant set S, the count k^*, the index m and the rank j. The continuous objects are wealth, c, V, μ, the draws, the scores and the functions u, κ, F, G, φ, H_m and Δ; the wealth law Λ and the laws of r and s enter as distributions. The utility of wealth u, the prize value V, and every object built from them (κ, Δ in its own units, U_i, and the sums of these across agents) are the modeller's constructs, attributed to the agents to rationalise recorded choices, rather than quantities in the world. The entry decision is a choice between lotteries, because the entry condition is exactly the comparison of the expected payoffs of entering and of not entering. The model is therefore of the von Neumann–Morgenstern kind in the sense of Schoemaker (1982). In a model of that kind utility "is used to represent preferences" rather than determining them (p. 532), and it is unique up to $aU+b$ with $a>0$ (p. 531). It is cardinal in the measurement sense only, since as a preference theory it "provides no more than ordinal rankings of lotteries" and "should not be interpreted as measuring strength of preference under certainty" (p. 533). Sen (1973) draws the matching line on the side of choice. Preference may be defined "as the underlying relation in terms of which individual choices can be explained", and it then "will simply be the binary representation of individual choice" (p. 253). Reading the same relation also as welfare is a further step, and "it is not in general possible to guarantee both simultaneously" (p. 259). Accordingly the entry condition is unchanged under $(u,V) \mapsto (au+b, aV)$ with $a>0$, and a statement about these constructs means something only if it survives that transformation. The last column separates five kinds of object, namely what an observer records, what is computed from records, facts of nature that exist but are never observed, constructs together with the recorded choices they rationalise, and computed references. Numerical example parameters (the CRRA coefficient, the witnesses, the constructions in the claim on burden-monotonicity) are not inputs and are omitted, as are the refuted claims R1 and R2 except where the manuscript reuses their objects.

Inputs

Table 1: Inputs of the model.

Symbol	Type	Represents	Obtained how
Q	Positive integer.	The number of challengers, "Q challengers plus one incumbent". A fact of the contest.	Recorded by the observer as the number of challengers in the contest.
C	A CDF.	The incumbent's score CDF. The incumbent is "scored as an entrant", $C=F$: the open contest, in which the holder competes again under the rule that admits the challengers (manuscript, Appendix D). Rows that hold for any atomless C say so. Her entry is a parameter in M1 to M29 and a choice in M30, where she is player 0 with the lowest cost.	Not recorded. $C=F$ is the manuscript's reading of the rule, and in M30 it is the equilibrium outcome whenever identities are pinned. A protected holder, whose C does not move with μ, is outside the model.

Symbol	Type	Represents	Obtained how
w_i; $(w_1,\ldots,w_Q)$; $w_{(j)}$, j	Real number, $w_i > c$; a vector in $\mathbb{R}^Q$; its descending order statistics; integer rank, 1 richest.	"Challenger i's resource." The realised profile is common knowledge among the challengers. A fact about each challenger, known to all of them.	Recorded by the observer before the entry decisions. Order statistics and ranks are computed from the record. The record is the observer's, whereas the challengers' knowledge of the profile is the manuscript's assumption and is not shown by the record.
Λ	A distribution across challengers, support bounded below by c.	"The law from which the profile is drawn." A fact of nature.	Never observed as a law; each recorded profile is a draw from it. It enters the results only through the realised order statistics, the mean $\bar{\mu}$, the value $1-\Lambda(\bar{\mu})$ and quantile functions.
c	Real number, $c > 0$, identical for all agents.	"The nominal cost, identical for all agents", paid "to acquire social capital". In resource units, and what society gives up.	Recorded by the observer as the outlay required to enter.
u	Function of wealth, increasing, satisfying burden-monotonicity; a cardinal index, fixed only up to $u \mapsto au+b$, $a>0$, together with $V \mapsto aV$.	"The utility of wealth." The modeller's construct, attributed to the agent, rather than a quantity in the world.	Not observed, being a construct. It rationalises the recorded entry decisions, and enters the model only through κ.
$\kappa(w) = u(w)-u(w-c)$	Function on (c,∞); continuous, strictly decreasing, $\kappa \to \infty$ as $w \downarrow c$; scales with a under the transformation above.	"The burden of c in utility units", the entry cost (utility). "The sole route by which the wealth distribution affects behaviour." The modeller's construct, attributed to the agent. It is a difference of a utility that represents preferences over lotteries, rather than a measure of how strongly the cost is felt (Schoemaker, 1982, pp. 533, 535).	Not observed, being a construct. It rationalises the recorded entry decisions. Only the ratio κ/V survives the transformation, and recorded entry decisions restrict that ratio to a set rather than to a value (Recovery 4).
V	Real number, $V > 0$, in the units of u; scales with a under the transformation above.	The prize's value. The prize is "exogenous and fixed", rank-allocated, a non-market good, additively separable and "valued identically by rich and poor". The modeller's construct, attributed to the agents.	Not observed, being a construct. It enters only as a scale factor of Δ (M1). The prize itself is recorded. Where two prizes are compared (M11), the recorded datum is the agents' choice between them, or their answer to a question ranking them, made before entry. An ordering of V is what rationalises either record. Sen (1973), pp. 257–258, counts answers to questions as information alongside behaviour, noting "well-known limitations of the questionnaire method".
μ	Real number; the manuscript works with $\mu \in (0,1)$ and the limit $\mu \to 0$.	The weight on the base component r in the score; $1-\mu$ weighs the bought component. A fact of the contest, set by the institution that runs it. The manuscript's Research question motivates it as that institution's trade-off: the bought component is a token it can verify cheaply, where the quality it cares about is costly to verify. The model takes μ as given and does not derive that choice.	Never observed as a number; enters only through F and G. Where the rule is published, the observer reads the value the token earns against the rest of the index (in Chiba City's daycare selection rules, four points for 64 or more hours a month of unlicensed care; in Italian tenders, the extra points each authority sets for a gender-equality certificate; sources in the manuscript's record `lit/applications_2026`), which orders rules by μ without measuring it. Conjecture R1 of the manuscript's Appendix E uses $\sup \operatorname{supp}(G) = \mu$, which holds when the support of r has upper endpoint 1, a condition the primitives do not state.

Symbol	Type	Represents	Obtained how
r_i	Real-valued draw; independent across players, of wealth and of s_i; arbitrary distribution.	The base component: the part of the score every challenger has, whether or not she enters. A fact of nature.	Never observed separately. For an outsider the recorded score is μr_i.
s_i	Real-valued draw, $s_i \geq 0$; independent across players, of wealth and of r_i; arbitrary distribution.	The bought component: the part of the score that entry adds. A fact of nature.	Never observed separately. For an entrant the recorded score is $\mu r_i + (1-\mu)s_i$.
F, G	CDFs; continuous, common lower endpoint; $F \leq G$ pointwise (M2).	"Law of σ for an entrant (F) and an outsider (G)." Facts of nature. "All results are stated in F and G only."	Computed from records as the distributions of recorded scores among entrants and among outsiders, pooled across challengers. Pooling across wealth uses the independence of the draws from wealth (M26 (anonymity)).
$\varphi = G - F$	Function, $\varphi \geq 0$, vanishing at both endpoints of the union of supports.	"The wedge entry buys."	Computed only, from F and G.
m	Integer in $\{0, \ldots, Q-1\}$.	The number of other challengers who enter, as faced by a challenger deciding. The rank-j challenger, entering as the j-th entrant, faces $m = j-1$.	Computed from the rank.
θ	Real number; $\theta = 0$ is the model.	Tilt of the law of s against wealth rank, outside the model's scope, used in the manuscript's Appendix D to show where count invariance fails.	Never observed as a parameter. The independence that θ relaxes is a direct comparison of recorded entrant scores across wealth ranks.
$w \mapsto w'$; T	A map between wealth profiles; T increasing for a pivot-spread.	"Any map between wealth profiles", the change in the wealth distribution that a comparative static is about.	Recorded as two wealth profiles, before and after a change, or of two contests. T is seen only at the recorded wealths.
ordering in dispersion	A partial order on wealth laws.	For quantile functions y and z, the second law is more dispersed when $z - y$ is non-decreasing (Lewis and Thompson, 1981); with a crossing point the transport map between the two laws is a pivot-spread, so M13 applies.	Computed from two recorded profiles as the sign pattern of the rank differences $w'_{(j)} - w_{(j)}$ along the ranks.
x_0	Real number, in wealth units.	The pivot, "the wealth at which the displacement changes sign."	Computed from two recorded profiles. In general it is located only between two adjacent ranks; in the linear family it is given by Recovery 1.
λ	Real number, $\lambda \geq 1$ (M13, M15).	The factor of the linear spread $T_\lambda(w) = x_0 + \lambda(w - x_0)$.	Computed from two recorded profiles (Recoveries 1–2).
$\bar{\mu}$	Real number, $\bar{\mu} = \mathbb{E}[w]$.	Mean wealth, and the pivot of the linear mean-preserving spread of M13 (the linear spread).	Computed from recorded wealths (Recovery 2).
Λ_α, Λ_β; r_j	Two wealth distributions; quantile levels.	The quantile coupling, which reads the displacement as $\Lambda_\beta^{-1}(r_j) - \Lambda_\alpha^{-1}(r_j)$.	The laws are never observed; the displacement is computed from two recorded sorted profiles. Here r_j is a quantile level rather than the draw r_i.

Outputs

Table 2: Outputs of the model.

Symbol	Type	Represents	Obtained how
e_i	Discrete, $\{0,1\}$.	The challenger's decision to enter, the only choice in the model.	Recorded by the observer as whether the challenger paid c, after the profile is known and before the draws.
S	Subset of $\{1,\ldots,Q\}$.	The entrant set, $S=\{i : e_i = 1\}$.	Recorded. One play shows the one equilibrium set played, and not the other sets the game admits.
k^*	Integer in $\{0,\ldots,Q\}$.	The entry count, the largest k with the entry condition holding at every rank $j \le k$. By M6 (count invariance) it is the size of every pure-strategy equilibrium entrant set.	Recorded as $\lvert S\rvert$. Its definition runs through κ and Δ, which are constructs, and the recorded count is what that definition rationalises.
$w_{(k^*)}$	Real number.	The marginal entrant, the rank-k^* challenger, and her wealth.	Computed from the recorded profile and recorded k^* as the wealth of the k^*-th richest challenger, whether or not she is in the recorded S. Under identity multiplicity she can be absent.
$\{(1),\ldots,(k^*)\}$	Subset of size k^*.	The assortative set, the k^* richest, equivalently the k^* with the lowest κ.	Computed from the recorded profile and k^*, and compared directly with the recorded S.
$\kappa(w_{(k^*+1)}) > \Delta(k^*-1)$	An inequality.	The condition of M7 (identities pinned), under which the assortative set is the unique equilibrium.	Not observed, being a statement about constructs. In the ratio form $\kappa(w_{(k^*+1)})/V > \Delta(k^*-1)/V$ it survives the transformation of u and V.
σ_i	Real number.	The score, "what the prize is awarded on."	Recorded after the draws.
$\mathbf{1}\{i \text{ wins}\}$	Discrete, $\{0,1\}$.	Who takes the prize, the holder of the highest score.	Recorded.
U_i	Real number, in the units of u.	The payoff, $u(w_i - ce_i) + V \cdot \mathbf{1}\{i \text{ wins}\}$. The modeller's construct, attributed to the agent.	Not observed, being a construct. It rationalises the recorded entry decision. Its level is not fixed by the transformation, since b is free.
H_m; M_m	A CDF; a random variable.	The opponent maximum, the "CDF of the best rival score when m other challengers enter". M_m is that best rival score.	$H_m = F^m G^{Q-1-m} C$ is computed from F, G, C, Q, m. Realisations of M_m are recorded as the best rival score in a contest with m other entrants.
$\Delta(m)$	Real number ≥ 0, in the units of u; a non-increasing function of m (M4 (monotonicity)).	The gain from entering, the right-hand side of the entry condition. A function of the entrant count alone (M26 (anonymity)).	$\Delta(m)$ itself is a construct, since it carries V. The ratio $\Delta(m)/V = \int H_m\,dF - \int H_m\,dG = \mathbb{E}[\varphi(M_m)]$ (M1) carries no utility and is computed from the recorded F, G, C, Q.
$\Delta(m+1) - \Delta(m)$; K	Real number ≤ 0; $K = F^m G^{Q-2-m} C$, a CDF product.	The step of the gain schedule, equal to $-\frac{V}{2}\int \varphi^2\,dK$ (M3 (the step identity)).	Computed only, as a multiple of V, from F, G, C, Q.
$\sum_{i\in S}\kappa(w_i)$; k^*c	Real number, in the units of u; real number, resource units.	Total entry cost in utility units, and its resource counterpart, which is identical across equilibria.	The utility total is a construct, and summing it across agents is the interpersonal comparison the manuscript's terminology flags (M8). k^*c is computed from the recorded k^* and c.
$\sum_i U_i$	Real number, in the units of u.	The unweighted sum of payoffs by which M8 (assortative selection) ranks equilibria.	Not observed, being a construct summed across agents, an interpersonal comparison the manuscript's terminology flags (M8). The payoffs summed are those that rationalise the entry choices. Reading their sum as welfare is a second link, which Sen (1973), pp. 254 and 259, says cannot in general be kept together with the first.

Symbol	Type	Represents	Obtained how
$\Delta(m) \to V/(m+2)$ as $\mu \to 0$	A function of m, independent of Q.	The M9 benchmark, with the incumbent scored as an entrant. The manuscript's row M9 carries the factor V.	A computed reference. It is a limit, not reached at any recorded $\mu > 0$.
$\Delta(m) \leq V/(m+1)$; $k^* \leq V/\kappa$	Bounds, uniform in Q.	M10, saturation.	Statements about constructs. In ratio form, $\Delta(m)/V \leq 1/(m+1)$ is a computed reference from the recorded F, G, C, Q, whereas $k^*\kappa/V \leq 1$ is not observed.
Direction of k^* in V and c	Sign of a weak change.	M11 states that k^* is non-decreasing in V and non-increasing in c.	Direct comparison of recorded k^* across contests that differ only in the recorded c, or only in the prize, with the prizes ordered by the agents' recorded choice between them (the V row of the Inputs table).
Direction of k^* in μ	Sign of a weak change.	M28 states that, with the incumbent scored as an entrant, k^* is non-increasing in μ for every atomless law of r and every law of $s \geq 0$.	μ is never recorded (the μ row of the Inputs table), so the comparison is across contests whose rules the observer orders by the weight they give to the base component, with c, the prize and the laws of r and s unchanged. The ordering of the rules is the observer's reading of the rules, not a measurement, and the direction is the only prediction.
Entrants' share of the prize	Real number in $[0,1]$.	M31: the probability $\xi(k^*)$ that the prize goes to an entrant, the incumbent included; non-decreasing in the count, $\frac{k+1}{Q+1}$ with k entrants when entry buys nothing, and non-increasing in μ.	Recorded directly: whether the winner paid c, across contests run under one rule. Its frequency estimates $\xi(k^*)$. No construct enters, unlike every quantity built on u.
Direction of k^* under a change of profile	Sign of a weak change, rise or fall.	M13 (the linear spread), M13 (pivot-spreads), Proposition (endogenous margin), M12 (margin condition).	Direct comparison of recorded k^* under two recorded profiles, with Q, c, the prize and the score technology unchanged. The hypothesis ($w_{(k^*)}$ against x_0, or the margin condition, which reads d_{k^*} for the rise branch and d_{k^*+1} for the fall branch) is computed from the two profiles and the pre-change k^*. No computed reference is needed.
d_j; the band	Real numbers; a yes-or-no property of the margin, holding when $d_{k^*} < 0 < d_{k^*+1}$.	$d_j = w'_{(j)} - w_{(j)}$, rank 1 richest. The margin condition is silent exactly inside the band, a pivot-spread never reaches the band, and inside the band the displacement signs determine nothing, for mean-preserving spreads too (M14).	Computed from two recorded sorted profiles and the recorded pre-change k^*. Only d_{k^*} and d_{k^*+1} are needed.
$1 - \Lambda(\bar{\mu})$	Real number in $[0,1]$.	The entry rate at which the sign of M13 (the linear spread) flips.	Computed from recorded wealths, and compared with the recorded entry rate.
$k^*(\lambda)$; $\bar{\lambda}$	A step function of λ; a real number.	M15 (strictness) states that k^* is non-increasing in λ, and strictly below $k^*(1)$ for $\lambda > \bar{\lambda}$.	$k^*(\lambda)$ is recorded only at the profiles observed. $\bar{\lambda}$ is not observed, since the manuscript asserts its existence only, through κ and $\Delta(k^*-1)$.
$\Delta_Q(0)$; $D(Q) = \Delta_{Q+1}(0) - \Delta_Q(0)$; W_Q	Real numbers; $W_Q = C\,G^{Q-1}(1-G)$, a function.	The first entrant's gain among Q challengers and its step in Q. M16 states that $D(Q) = -V \int W_Q\,(dF - dG)$, for every incumbent law C, and that $\Delta_Q(0)$ rises in Q iff $\int W_Q\,dF < \int W_Q\,dG$.	A computed reference, from the recorded F, G and Q. It is the change in the gain schedule rather than the recorded change in k^* across contests of different Q.
μ^*	Real number; distribution-dependent.	No row of the manuscript defines it. An earlier draft reported a value of μ at which the sign of $D(Q)$ reverses, found by quadrature, which is an illustration only. The manuscript shows that the sign depends on the law of r (M16 to M22).	Not used.

Structure

- *Entering only through others.* u enters only through κ; wealth enters only through $\kappa(w_{(j)})$ (M26 (anonymity)); μ and the laws of r and s enter only through F and G; V enters only as a scale factor of Δ, and the entry condition is unchanged under $(u,V) \mapsto (au+b, aV)$, $a>0$; Λ enters only through the realised profile, $\bar{\mu}$, $1-\Lambda(\bar{\mu})$ and quantile functions.
- *Distributions.* Λ across challengers, and the laws of r and s, which reach the results only as F and G.
- *Needing a computed reference.* Δ/V and its step, the M9 benchmark, the first M10 bound in ratio form, the M16 sign, μ^*, and the flip entry rate $1-\Lambda(\bar{\mu})$.
- *Given directly by recorded comparisons.* $F \leq G$ (recorded scores of entrants against outsiders); independence of the draws from wealth (recorded scores across wealth ranks, within one entry status); the recorded S against the assortative set; the direction of k^* across contests differing only in c; the direction of k^* across two recorded profiles, with its hypothesis checked on the same records.
- *Facts that exist but are never observed.* μ, r_i, s_i, and Λ as a law.
- *Constructs, not observed.* u, κ, V, Δ in its own units, U_i, $\sum_i U_i$, $\sum_{i\in S}\kappa(w_i)$, the condition of M7 (identities pinned), the second M10 bound, and $\bar{\lambda}$. Each rationalises recorded entry decisions, or a recorded choice between prizes. Under $(u,V) \mapsto (au+b, aV)$ the quantities that do not change are exactly the functions of κ/V (and of Δ/V, which carries no utility). U_i and $\sum_i U_i$ move with a and b, whereas $\sum_{i\in S}\kappa(w_i)$ moves with a only. The difference of either sum between two equilibria keeps its sign. Summing across agents is the interpersonal comparison the manuscript's terminology flags (M8).

Recovery formulas

1. *Linear family* $T_\lambda(w) = x_0 + \lambda(w - x_0)$. From two recorded sorted profiles, take ranks a, b with $d_a = w'_{(a)} - w_{(a)}$ and $d_b = w'_{(b)} - w_{(b)}$. Then

$$\lambda = 1 + \frac{d_a - d_b}{w_{(a)} - w_{(b)}}, \qquad x_0 = \frac{d_b\, w_{(a)} - d_a\, w_{(b)}}{d_b - d_a}.$$

   The formula for λ needs $w_{(a)} \neq w_{(b)}$. The formula for x_0 needs $d_a \neq d_b$, equivalently $\lambda \neq 1$ and $w_{(a)} \neq w_{(b)}$. Both use four recorded wealths.

2. *Linear mean-preserving spread of M13 (the linear spread).* $\bar{\mu}$ is the mean of the recorded profile, and the spread about $\bar{\mu}$ preserves that mean. Then $\lambda = (w'_{(j)} - \bar{\mu})/(w_{(j)} - \bar{\mu})$ for any rank j with $w_{(j)} \neq \bar{\mu}$. The formula uses one recorded profile and one rank of the other.

3. *Gain schedule, up to* V. $\Delta(m)/V = \int H_m\, dF - \int H_m\, dG = \mathbb{E}[\varphi(M_m)]$, from the recorded F, G, C and Q (M1). The formula needs the regularity of the primitives (F, G continuous, common lower endpoint).

4. *Ratios that rationalise a recorded equilibrium, as a set rather than a value.* For a recorded equilibrium S with $|S| = k$, $\kappa(w_i)/V \leq \Delta(k-1)/V$ for $i \in S$ (when $k \geq 1$), and $\kappa(w_i)/V > \Delta(k)/V$ for $i \notin S$ (when $k \leq Q-1$). These are the equilibrium conditions of M6 (count invariance) divided by $V > 0$, with Δ/V from Recovery 3.

References

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